SESSION 2

Becoming a GeoAI Innovator

Innovator Traits · The Innovation Process · Design Thinking in Depth · Problem-Solving Mindset · Stakeholder Engagement

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

How to use these notes. *These notes are self-contained: they cover everything from the live session plus expanded explanations of the Design Thinking toolkit, a full worked example, diagrams, a glossary, and self-check questions.*

Learning Objectives

By the end of this session, you should be able to:

- Name the six characteristics that distinguish successful GeoAI innovators, and the real innovators who embody them.
- Apply the five-step Innovation Process (Observe → Question → Ideate → Prototype → Learn) to a real problem.
- Map the Innovation Process onto the Stanford school Design Thinking framework and use its concrete tools — empathy maps, POV statements, "How Might We" questions, low-fidelity prototypes, and think-aloud testing.
- Use four rapid problem-solving techniques and four rapid stakeholder-engagement techniques under hackathon time pressure.
- Produce a Stakeholder–Pain Point Matrix and a one-sentence problem definition (Activity 2).

1. Characteristics of GeoAI Innovators

Trait	What it looks like in practice
Curious	Asks why the data looks the way it does before jumping to a model.
Interdisciplinary	Comfortable bridging geography, data science, and social context.
Systems Thinker	Sees how a solution ripples across communities, land, and policy.
User-Centred	Designs with — not just for — the people who will use the solution.
Ethically Grounded	Weighs risk and fairness as carefully as accuracy and speed.
Resilient	Treats failed prototypes as data, not defeat.

Innovators to Know — the Traits at Global Scale

- **Jack Dangermond (co-founder & president, Esri):** built ArcGIS from a 1969 land-use consultancy into the world's leading GIS platform, now integrating AI/ML tools used by over 700,000 organisations worldwide, including most national governments.
- **Rebecca Moore (founder, Google Earth Engine):** turned a local map that stopped a redwood-logging plan into a planetary-scale cloud platform behind Global Forest Watch, now used by 500,000+ scientists and organisations.
- **Will Marshall (co-founder & CEO, Planet Labs):** co-founded Planet in 2010 to image the entire Earth's landmass daily with small satellites — now a 200+ satellite fleet feeding AI models for agriculture, disaster response, and security.

- **The GeoAI research community:** the ACM SIGSPATIAL GeoAI Workshop coined the term "GeoAI" at its first meeting in 2017 and has grown into the field's central academic venue, uniting geographers, computer scientists, and industry.

Notice that none of these stories began with technology for its own sake — each began with a problem someone cared about (land use, a threatened forest, blind spots in Earth observation) and scaled from there.

2. The Innovation Process — a Repeatable Cycle

Innovation is not a single flash of insight; it is a repeatable five-step cycle. Insights from "Learn" loop back into fresh "Observe" and "Question" rounds throughout the hackathon.

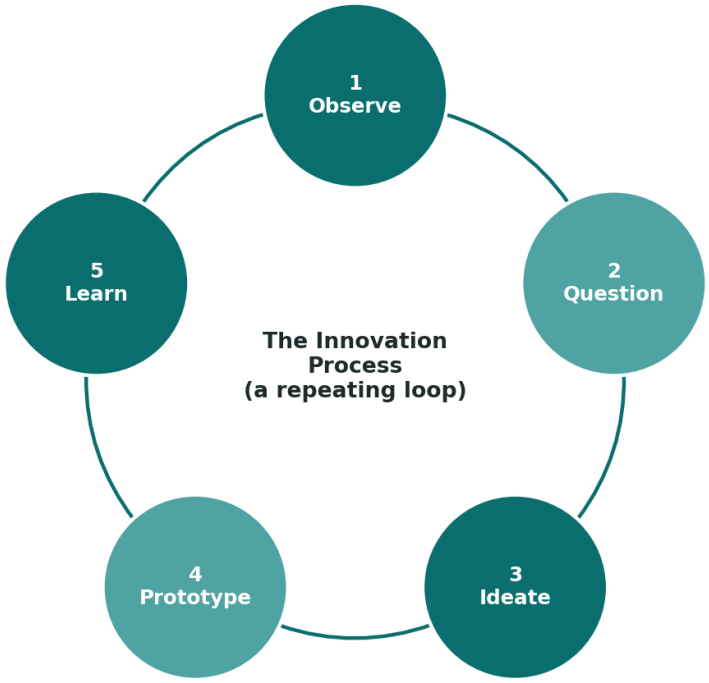


Figure 2.1 — The Innovation Process is a loop: what you learn feeds the next round of observation.

***Running worked example:** "Team Tropika" — a hackathon team tackling flash floods in a Klang Valley neighbourhood. Follow them through every step below; the same team returns in Session 5 for the ethics walkthrough.*

Step 1 · Observe

Go to where the problem actually happens and watch, without jumping to solutions. Look for what people do, work around, or complain about — the gap between what's designed and what's real.

Team Tropika: visiting a low-lying neighbourhood after a storm, the team notices residents rely on word-of-mouth and a community WhatsApp group to know when water is rising — not any official channel.

Step 2 · Question

Turn raw observations into sharp, testable questions that expose assumptions and point at root causes, not symptoms.

Team Tropika: "Why does the official warning arrive after residents already know from neighbours?" Investigation reveals river-gauge readings update only every few hours — too slow for a catchment that floods in under 90 minutes.

Step 3 · Ideate

Generate a wide spread of possible responses before judging any. Quantity first, quality later — the best idea is rarely the first one written down.

Team Tropika: options listed: a denser sensor network, an SMS alert service, a crowdsourced water-level app, or an ML model forecasting risk from rainfall + existing telemetry. The ML model is shortlisted as fastest to prototype with open data.

Step 4 · Prototype

Build the smallest, cheapest version that can be tested — not the finished product. A prototype exists to answer a question, so build only enough to answer it.

Team Tropika: in one sprint the team trains a simple model on public rainfall and river-level data for one catchment, producing a 6-hour flood-risk score, plus a clickable WhatsApp-bot mockup — no real backend.

Step 5 · Learn

Put the prototype in front of real users or real data and honestly capture what breaks. Learning is not optional polish — it is the whole point of prototyping, and it feeds directly back into Observe.

Team Tropika: tested against a past storm, the model over-warns during minor rain (too many false alarms), and residents say they would ignore an SMS but trust a WhatsApp voice note. Both insights reshape the next iteration.

Team Tropika's actual prototype: three sliders simulating rainfall, river level and soil saturation; a simple weighted formula (not yet a trained model) producing a live risk score and gauge; a "Run Prototype" button generating the exact WhatsApp-style alert at each risk tier; and a one-click "Heavy Monsoon Scenario" that reproduces the Learn-phase test. A real, clickable file — deliberately not a polished product. That is what "cheapest testable version" means.

3. Design Thinking — the How Behind the What

The Innovation Process describes what good innovators do. Design Thinking supplies the how — a well-tested, human-centred methodology developed and popularised by the Hasso Plattner Institute of Design at Stanford (the "d.school") and the design firm IDEO, with concrete tools for every stage, so the whole team works from one shared playbook instead of improvising under time pressure. It is used by organisations from Apple and Google to IBM, and its five stages map one-to-one onto the Innovation Process:

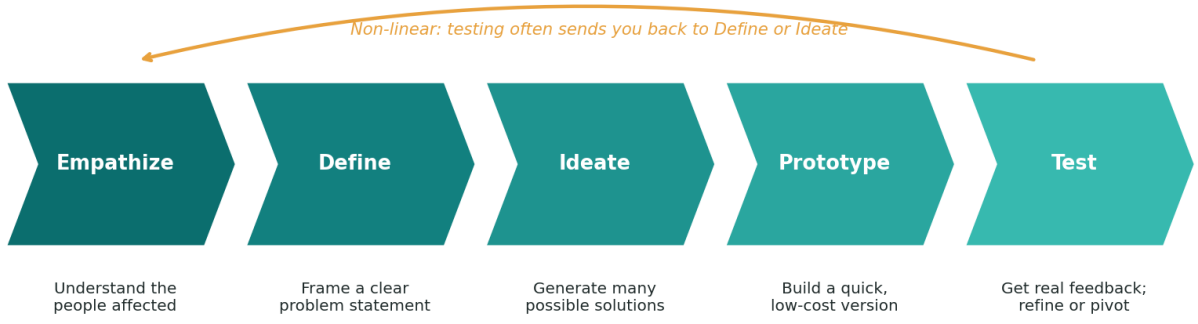


Figure 2.2 — The five Design Thinking stages. The process is explicitly non-linear: testing routinely sends teams back to Define or Ideate.

Innovation Process	Design Thinking stage	What Design Thinking adds
Observe	Empathize	Structured empathy tools — interviews, contextual observation, empathy maps.

Innovation Process	Design Thinking stage	What Design Thinking adds
Question	Define	A crisp problem statement and a “How Might We” question the team can rally around.
Ideate	Ideate	Structured brainstorming — Crazy 8s, idea clustering, dot-voting to narrow fairly.
Prototype	Prototype	Rapid, low-fidelity building — paper prototypes, clickable mockups.
Learn	Test	Structured feedback capture — think-aloud testing, explicit triggers to iterate again.

The Five Stages, In Depth (with Tools)

Stage 1 · Empathize. Understand the people affected — their needs, context, and constraints — by doing research rather than assuming. Core tools: user interviews (ask people to show you how they do a task, and why), contextual observation (watch the real environment, not a description of it), and the empathy map — a canvas divided into four quadrants capturing what a user Says (direct quotes), Thinks (unspoken beliefs and worries), Feels (emotions), and Does (observed behaviour). The d.school stresses that our minds automatically filter out information, so empathy work exists to help us see with fresh eyes.

Stage 2 · Define. Synthesise what you learned into a clear, human-centred problem statement — often written as a Point-of-View (POV): “[User] needs a way to [need] because [insight].” Then convert it to a “How Might We” (HMW) question that opens the door to many solutions. Calibrate the scope: “How can we improve the whole flood experience?” is too broad to act on; “How can we redesign one alert button?” is too narrow to inspire. Aim between.

Stage 3 · Ideate. Generate a wide range of possible solutions without judging them yet — quantity first, refinement later. Tools: classic brainstorming (no criticism during generation), mind mapping (visualising connections between ideas), Crazy 8s (8 sketched ideas in 8 minutes), Worst Possible Idea (deliberately propose terrible ideas, then reverse-engineer the insight inside them), and dot-voting to narrow the field fairly across the team.

Stage 4 · Prototype. Build a quick, low-cost version of your best idea to make it tangible. Start low-fidelity (paper sketches, wireframes, cardboard mockups) and only increase fidelity once the concept survives feedback. d.school rules of thumb: don’t fall in love with one prototype — let go early; and identify the variable — each prototype should exist to answer one specific question.

Stage 5 · Test. Put the prototype in front of real (or proxy) users and watch. Use think-aloud testing — ask the user to narrate what they are doing and expecting as they use it — and show, don’t tell, so their reactions are genuine. Failed tests are not the end: they send you back to Define or Ideate with better information, which is exactly how the process is meant to work.

Why Design Thinking, Why Now?

- **Stops the #1 hackathon failure:** brilliant tech, wrong problem — Empathize and Define come before any building.
- **One shared playbook:** a common vocabulary and toolkit across every discipline on the team.
- **Humans stay central:** every decision starts and ends with the people affected — priming Session 3’s ethics focus.
- **Concrete deliverables:** turns five abstract steps into things you can hand in — empathy map, POV statement, HMW question, prototype, test script.

4. Problem-Solving Mindset — Why & How

Why it matters. In a time-boxed hackathon the real risk isn't running out of time to build — it's spending that time building the wrong thing. Technical teams especially fall into solution-first thinking: falling for a tool or model, then searching backward for a problem it fits.

Four techniques, under an hour combined:

Technique	Time	What it does	Team Tropika example
"5 Whys"	~10 min	Ask "why" five times on your first problem statement to reach the root cause, not a symptom.	"Warnings arrive late" → the real issue is gauge data updating only every few hours.
"How Might We" reframe	~15 min	Turn a complaint into an open question that invites many solutions instead of locking in one.	"Warnings are too slow" becomes "How might we cut the time between rising water and a phone buzzing?"
Defer-judgement brainstorm	~20 min	Separate generating ideas from evaluating them — judging too early kills good-but-unusual ideas.	12 ideas listed before judging any; a "silly" community-radio idea sparks the SMS-alert concept that survives.
Kill-criteria check	~5 min	Before building, write down what would prove the idea wrong — then go check first.	"Kill this if free rainfall data doesn't cover our catchment" — checked first, saving a wasted afternoon.

Core habits behind all four: reframe problems before rushing to solutions; break big challenges into testable pieces; stay comfortable with ambiguity and iteration; and separate the person from the idea when giving feedback.

5. Stakeholder Engagement — Why & How

Why it matters. GeoAI decisions land on real places and real people — a flood alert that's too slow, a crop model trained on the wrong region. Solutions built without the people affected often fail at adoption even when the technology works — and judges score projects on exactly this: who did you engage, and what changed because of it?

Four check-ins, under an hour combined:

Technique	Time	What it does	Team Tropika example
Power / interest map	~10 min	Plot who's affected against who can act, starting from the Stakeholder field on your GeoAI Canvas.	Residents (high interest, low power) vs. DID/JPS (high power, high interest) — the one interview slot goes to DID.
Proxy interviews / role-play	~20 min	Can't reach real residents this weekend? Interview someone one step removed, or role-play the persona.	The team interviews a community WhatsApp group admin instead of flood victims directly.
Borrow existing voices	~15 min	News reports, WhatsApp groups, complaint logs and past studies often already contain what stakeholders have said.	Pulling complaint threads from a public flood Facebook group instead of running a new survey.

Technique	Time	What it does	Team Tropika example
Rapid validation loop	~10 min	Show your prototype to one real or proxy stakeholder before the final push — a 5-minute reaction beats a wrong guess.	The group admin says the alert text is too long to read in an emergency, so it's shortened before the demo.

Core habits: identify who is impacted, directly and indirectly; involve affected communities early, not just at launch; listen for pain points data alone won't reveal; validate assumptions with the people who live them.

6. Hands-On Activity 2 — Design Thinking Exercise (12 minutes)

Challenge: the Urban Heat Island problem (see Session 1, Case Study 3.2 for the KL background). In teams — or on paper if self-studying — analyse the challenge, identify the stakeholders, their pain points, and the outcomes they would want to see.

1. List everyone affected by urban heat — go past "residents" to specifics: outdoor construction workers, elderly residents without air-conditioning, school children, emergency services.
2. Identify each group's specific pain points.
3. Define the problem in one clear sentence.

Deliverables: a Stakeholder–Pain Point Matrix and a Problem Definition Sheet.

Stakeholder group	Pain point	Desired outcome
e.g. Outdoor construction workers	e.g. Heat exhaustion risk during peak hours	e.g. Real-time heat-risk alerts for work sites
Your answer:		

Problem definition template: "How might we help _____ so that _____?"

Glossary

Term	Meaning
Design Thinking	A human-centred, iterative problem-solving methodology (Stanford d.school / IDEO) with five stages: Empathize, Define, Ideate, Prototype, Test.
Empathy map	A four-quadrant canvas capturing what a user Says, Thinks, Feels, and Does.
POV statement	Point-of-View problem statement: "[User] needs a way to [need] because [insight]."
"How Might We" (HMW)	An open question format that reframes a problem to invite many solutions.
Crazy 8s	An ideation exercise: sketch 8 different ideas in 8 minutes.
Dot-voting	Each team member places dots on their preferred ideas — a fast, fair way to converge.
Low-fidelity prototype	A rough, cheap version (paper sketch, clickable mockup) built to test one question quickly.
Think-aloud testing	A test method where users narrate their thoughts while using a prototype.
T-shaped teammate	Deep expertise in one skill, working fluency across many.

Term	Meaning
Kill criteria	Pre-written conditions that would prove an idea wrong — checked before building.

Self-Check Questions

1. Write the Team Tropika insight from the Observe step as a formal POV statement.
2. The team's first problem statement was "warnings are too slow." Apply two rounds of "why?" — what root cause do you reach?
3. Why does Design Thinking insist on deferring judgement during Ideate? What is lost when you evaluate ideas as they are generated?
4. Give one reason a low-fidelity prototype can be more useful than a polished one in the Test stage.
5. Your team cannot reach any real flood victims this weekend. List two legitimate substitutes from the stakeholder-engagement toolkit.

Key Takeaways

- GeoAI innovators are curious, interdisciplinary, systems-thinking, user-centred, ethically grounded, and resilient — traits visible in real figures like Dangermond, Moore, and Marshall.
- Innovation is a repeatable cycle — Observe, Question, Ideate, Prototype, Learn — not a single flash of insight.
- Design Thinking turns that cycle into a shared playbook with concrete tools (empathy maps, POV, HMW, Crazy 8s, think-aloud tests) and deliverables at every stage — and it is explicitly non-linear.
- An hour of structured problem-framing (5 Whys, HMW, defer-judgement, kill criteria) can save a full day of building the wrong thing.
- Four short stakeholder check-ins are usually enough to catch the one assumption that would sink the whole project.

Sources & Further Reading

- Stanford d.school — An Introduction to Design Thinking: Process Guide (the original five-mode guide).
- Interaction Design Foundation — The 5 Stages in the Design Thinking Process (ixdf.org).
- IDEO.org — Design Kit; IDEO — Human-Centred Design Toolkit.
- Esri; Google Sustainability — Global Forest Watch; Planet Labs; ACM SIGSPATIAL GeoAI Workshop series (2017–).